

Lab 4- Part 2

THEORY: What is a ComboBox?

A ComboBox is a dropdown list. User clicks it and sees options to choose from.

Real life example: When you fill a form online and click on "Country" or "City", a list drops down showing options like Dhaka, Saidpur, etc.

STEP 1: Simplest ComboBox with One Item

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.ComboBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        ComboBox<String> locationCombo = new ComboBox<>();
        locationCombo.getItems().add("Dhaka");

        VBox root = new VBox(10, locationCombo);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Step 1: One Item");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

What you see: A dropdown box showing "Dhaka"

Task: Change "Dhaka" to your city name, also add more options to the dropbox.

STEP 2: Add Multiple Items Using add() Method

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.ComboBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        ComboBox<String> locationCombo = new ComboBox<>();

        locationCombo.getItems().add("Dhaka");
        locationCombo.getItems().add("Saidpur");
        locationCombo.getItems().add("Chittagong");
        locationCombo.getItems().add("Rajshahi");

        VBox root = new VBox(10, locationCombo);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Step 2: Multiple Items");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

What you see: Click the dropdown. You see Dhaka, Saidpur, Chittagong, Rajshahi.

Task: Add two more cities.

STEP 3: Add Multiple Items Using addAll() Method (Easier)

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.ComboBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        ComboBox<String> locationCombo = new ComboBox<>();

        locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
        "Rajshahi", "Khulna");

        VBox root = new VBox(10, locationCombo);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Step 3: Using addAll");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

What you see: Same result but code is shorter.

Task: Add "Barishal" and "Sylhet" using addAll.

STEP 4: Add a Prompt Text (Hint for User)

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
```

```

import javafx.scene.control.ComboBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        ComboBox<String> locationCombo = new ComboBox<>();

        locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
"Rajshahi");
        locationCombo.setPromptText("Select your city");

        VBox root = new VBox(10, locationCombo);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Step 4: With Prompt Text");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}

```

What you see: Box shows "Select your city" in gray before user clicks.

Task: Change prompt text to "Choose location".

STEP 5: Set a Default Value

Code:

```

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.ComboBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

```

```

@Override
public void start(Stage primaryStage) {
    ComboBox<String> locationCombo = new ComboBox<>();

    locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
    "Rajshahi");
    locationCombo.setValue("Dhaka");

    VBox root = new VBox(10, locationCombo);
    Scene scene = new Scene(root, 300, 200);

    primaryStage.setTitle("Step 5: Default Value");
    primaryStage.setScene(scene);
    primaryStage.show();
}

public static void main(String[] args) {
    launch(args);
}
}

```

What you see: "Dhaka" is already selected when app opens.

Task: Change default value to "Saidpur".

STEP 6: Get Selected Value from ComboBox

Code:

```

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.Label;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {

```

```

        ComboBox<String> locationCombo = new ComboBox<>();
        locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
        "Rajshahi");
        locationCombo.setValue("Dhaka");

        Button showButton = new Button("Show Selected Location");
        Label resultLabel = new Label();

        showButton.setOnAction(e -> {
            String selectedLocation = locationCombo.getValue();
            resultLabel.setText("You selected: " + selectedLocation);
        });

        VBox root = new VBox(10, locationCombo, showButton, resultLabel);
        Scene scene = new Scene(root, 300, 250);

        primaryStage.setTitle("Step 6: Get Selected Value");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}

```

What you see:

- Select a city from dropdown
- Click button
- Label shows what you selected

Task: Change the button text to "Confirm Location".

STEP 7: Handle When Nothing is Selected

Code:

```

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;

```

```

import javafx.scene.control.Label;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        ComboBox<String> locationCombo = new ComboBox<>();
        locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
"Rajshahi");
        locationCombo.setPromptText("Select your city");

        Button showButton = new Button("Show Selected Location");
        Label resultLabel = new Label();

        showButton.setOnAction(e -> {
            String selectedLocation = locationCombo.getValue();

            if(selectedLocation == null) {
                resultLabel.setText("Please select a location first");
            } else {
                resultLabel.setText("You selected: " + selectedLocation);
            }
        });

        VBox root = new VBox(10, locationCombo, showButton, resultLabel);
        Scene scene = new Scene(root, 300, 250);

        primaryStage.setTitle("Step 7: Handle No Selection");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}

```

What you see: If you click button without selecting anything, it shows a warning message.

Task: Change the warning message to "Error: No city selected".

STEP 8: Style the ComboBox

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.ComboBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        ComboBox<String> locationCombo = new ComboBox<>();
        locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
        "Rajshahi");
        locationCombo.setValue("Dhaka");
        locationCombo.setStyle("-fx-font-size: 14px; -fx-background-color:
        lightblue;");

        VBox root = new VBox(10, locationCombo);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Step 8: Styled ComboBox");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Task: Change background color to light yellow and font size to 16px.

STEP 9: Add Label with ComboBox (Like Registration Form)

Code:


```

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.ComboBox;
import javafx.scene.control.Label;
import javafx.scene.layout.HBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label locLabel = new Label("Location:");
        ComboBox<String> locationCombo = new ComboBox<>();
        locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
"Rajshahi");
        locationCombo.setValue("Dhaka");

        HBox locBox = new HBox(10, locLabel, locationCombo);

        VBox root = new VBox(10, locBox);
        Scene scene = new Scene(root, 300, 100);

        primaryStage.setTitle("Step 9: Label with ComboBox");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}

```

What you see: Label "Location:" next to the dropdown box.

Task: Create another HBox for "Country" with Label and ComboBox.

STEP 10: Complete Location ComboBox Ready for Registration Form

Code:

```

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.Label;
import javafx.scene.layout.HBox;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label locLabel = new Label("Location:");
        ComboBox<String> locationCombo = new ComboBox<>();
        locationCombo.getItems().addAll("Dhaka", "Saidpur", "Chittagong",
"Rajshahi", "Khulna", "Barishal", "Sylhet");
        locationCombo.setValue("Dhaka");
        locationCombo.setPromptText("Select your city");

        HBox locBox = new HBox(10, locLabel, locationCombo);

        Button showButton = new Button("Show Selection");
        Label resultLabel = new Label();

        showButton.setOnAction(e -> {
            String selected = locationCombo.getValue();
            if(selected != null) {
                resultLabel.setText("Selected location: " + selected);
            } else {
                resultLabel.setText("No location selected");
            }
        });

        VBox root = new VBox(15, locBox, showButton, resultLabel);
        Scene scene = new Scene(root, 400, 200);

        primaryStage.setTitle("Location ComboBox - Ready for Registration Form");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {

```

```
        launch(args);  
    }  
}
```

What you see: Complete Location ComboBox with label, dropdown, button to show selection.

SUMMARY TABLE FOR COMBOBOX

Method	What it does	Example
<code>getItems().add("item")</code>	Adds one item	<code>locationCombo.getItems().add("Dhaka")</code>
<code>getItems().addAll("a","b")</code>	Adds multiple items	<code>locationCombo.getItems().addAll("Dhaka","Sa</code>
<code>setPromptText("text")</code>	Shows hint text	<code>locationCombo.setPromptText("Select city")</code>
<code>setValue("item")</code>	Sets default selection	<code>locationCombo.setValue("Dhaka")</code>
<code>getValue()</code>	Gets selected item	<code>String city = locationCombo.getValue()</code>
<code>setStyle("css")</code>	Changes appearance	<code>locationCombo.setStyle("-fx-font-size:14px;</code>

QUICK CHECK TASKS

1. Create a ComboBox for "Country" with 4 countries
 2. Set "Bangladesh" as default value
 3. Add a button that shows selected country in a Label
 4. Add prompt text "Choose your country"
 5. Style it with green background
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